

Causal Certificates for Proxy Discrimination: An Exact Bridge from Demographic Parity to Per-Decision Necessity and Sufficiency

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Abstract A fairness audit lives in two worlds that rarely meet. Group-fairness metrics such as demographic parity and equal opportunity are population averages, so they can say that a process is biased but cannot explain or justify any single decision. Per-instance explanations such as Shapley values and counterfactuals describe one decision, yet they carry no fairness guarantee and do not aggregate to any group quantity. Proxy discrimination, in which a model re-learns a withheld protected attribute A through correlated features ($A \rightarrow X \rightarrow \hat{Y}$), is exactly the phenomenon that needs both worlds at once, because it calls for a per-applicant answer to the question of whether someone was rejected because of a proxy, and that answer must be tied to the population disparity. We provide that bridge. Starting from the single local-accuracy property of additive explanations, we derive an exact per-feature decomposition of the demographic-parity gap, a localizer that recovers the proxy channel, and per-decision causal certificates that state the necessity, sufficiency, and causation of the proxy channel for an individual outcome through a directed, label-free counterfactual. Our central result is a consistency theorem showing that the individual causal effect averages back exactly to the per-cluster term of the group decomposition, so that the per-decision explanation and the population metric become two views of one quantity. We further show that the attribution must be interventional, because a conditional (observational) Shapley estimator spreads proxy credit onto correlated legitimate features and biases the split between direct and indirect effects, whereas the interventional estimator stays faithful while preserving the exact decomposition. On a controlled recruitment testbed with ground-truth proxies, localization is exact (average precision 1.00), the certificate finds that 15.7% of disadvantaged rejections are caused by the proxy channel, and the certificate stays conservative precisely when the channel also carries legitimate signal. The same decomposition powers a targeted repair that cuts the parity gap by 92% at no more than 0.02 AUC cost, and we report openly that minimizing parity can enlarge the equal-opportunity gap. On the real, published FairCVdb benchmark the audit transfers cleanly, localizing the demographic embedding with precision 1.00 and cutting the measured parity gap by 60% at an AUC cost of 0.014.

Keywords: algorithmic fairness, proxy discrimination, actual causation, necessity and sufficiency, Shapley values, causal mediation

1 Introduction

Consider an automated hiring screen that, by policy, is never shown the applicant’s gender. A regulator asks two questions. The first is about the *system*: “does this screen disadvantage women?” The second is about a *person*: “was *this* applicant rejected because of her gender?” Today these questions are answered by different, disconnected tools. Group-fairness metrics answer the first with a single number, a demographic-parity or equal-opportunity gap, but that number is a population average and says nothing about any individual file. Per-instance explainability answers the second with a Shapley attribution or a counterfactual, but those carry no fairness semantics and do not add up to the group gap. Neither tool, alone, can certify that a particular rejection was *caused* by a protected attribute the model was never given.

This disconnect is most acute for *proxy discrimination*. Two influential benchmarks make the mechanism vivid. *FairCVtest* [1, 2] builds a multimodal recruitment testbed in which a blind, competency-only score is corrupted by a group penalty, and shows that a model denied the protected attribute reconstructs it from correlated channels. *FairJob* [3], a million-row Criteo log, reports the same on real data: an “unaware” model matches an “unfair” one because the attribute is recovered from the feature vector. Both summarize the effect as a causal chain $A \rightarrow X \rightarrow \hat{Y}$.

Yet the diagnosis stops at the population: neither says *which* features carry the attribute, *how much* each contributes to the measured gap, and, most importantly, whether the chain is the *actual cause* of any individual’s outcome.

The gap between these two views is not merely technical. Anti-discrimination law distinguishes disparate *impact*, a population-level disparity, from disparate *treatment*, the use of a protected attribute in an individual decision [4], and a defensible audit must speak to both. A regulator who learns only that a screen rejects women at a higher rate cannot tell whether a given applicant has a grievance, and a developer who sees only a Shapley attribution for one applicant cannot tell whether the pattern is systemic. The two numbers can even point in opposite directions: a model may show a sizeable group gap while no single decision turns on the proxy, or conversely concentrate its harm in a few proxy-driven decisions that a population average dilutes to near-invisibility. What is missing is an object that is simultaneously a statement about a person and a statement about the population, with a guarantee that the two agree.

This paper. We close the loop between the population and the person. Our key observation is that one object, the additive (Shapley) decomposition of the model’s score, can be read at both scales as long as it is paired with the right counterfactual operator. We use a directed, label-free intervention that removes the protected direction inside a localized proxy channel, and we show that this single operator yields a complete account of proxy discrimination, from the group gap down to the individual decision.

Four questions guide the work, and each follows directly from a gap left open by the diagnostic benchmarks [1, 3] and the broader fairness-explanation literature [5]. The first concerns *localization*: among the many explanation methods available, which one most faithfully recovers the proxy channel that a model actually relies on, rather than features that are merely predictive or merely correlated with the protected attribute? The second concerns *attribution*: once a channel is found, how should per-feature credit be assigned so that the decomposition of the group disparity stays causally faithful instead of being smeared across correlated features? The third concerns *per-decision causation*: can we certify, for a single applicant, that the proxy channel was a genuine cause, both necessary and sufficient, of their outcome, and can that certificate stay cautious when the same channel also does legitimate work? The fourth concerns *consistency*: do the per-decision certificates add back up exactly to the population fairness metric, so that the individual view and the group view can never disagree? Our answers combine into a single recommended pipeline, namely PLS localization, interventional Shapley attribution, and a directed-counterfactual certificate, and we show that this combination outperforms the natural single-method baselines.

Contributions.

1. **An exact group decomposition (Proposition 1).** By local accuracy alone, the demographic-parity gap of the score equals the sum of per-feature terms $\Delta_j = \mathbb{E}[\phi_j \mid A=1] - \mathbb{E}[\phi_j \mid A=0]$, with no residual, giving an auditable ledger for the $A \rightarrow X \rightarrow \hat{Y}$ claim.
2. **Per-decision causal certificates (Definition 1).** Through a directed counterfactual we define the *necessity*, *sufficiency*, and *causation* of the proxy channel for an individual decision, importing the necessity/sufficiency calculus of actual causation [6, 7] into fairness auditing. A decision that is both necessary- and sufficient-attributable is *caused* by the proxy channel.
3. **A consistency theorem (Theorem 1).** The individual causal effect averages back *exactly* to the per-cluster term of the group decomposition. The per-decision certificate and the group metric are two projections of one quantity, which is the formal bridge between the regulator’s two questions.
4. **The certificate is honest under mixing.** When the proxy channel also carries legitimate signal, sufficiency drops and the causal rate falls *below* necessity: the certificate refuses to over-attribute, exactly when it should.
5. **Faithful attribution must be interventional.** Reading the gap as a natural indirect (proxy-path) plus direct effect, we show a conditional (observational) Shapley estimator smears proxy credit onto correlated legitimate features and so biases the split between direct and indirect effects, while the interventional estimator stays faithful, both preserving the exact decomposition.
6. **A targeted repair and an honest tension.** The same localization drives a surgical neutralizer that cuts the parity gap by 92% at ≤ 0.02 AUC cost on a ground-truth testbed, with A-direction orthogonalization preserving utility that deletion destroys; we show, not hide, that the repair can worsen equal opportunity.

We validate every claim with injected ground truth on a faithful FairCVtest-recipe testbed, confirm the audit on the real, published FairCVdb, and ship runnable transfer scripts plus an interactive auditor.

2 Related Work

Group fairness and unawareness. That deleting A does not remove its influence is foundational [8, 9]: redundant encodings survive. Parity is enforced by reductions [10], post-processing [11], or data repair [12]; all act on the whole feature vector and report population rates. We instead *attribute* the gap and act on the specific channel responsible.

Proxy and causal fairness. FairCVtest [1, 2] and FairJob [3] establish proxy leakage empirically; SensitiveNets [13] learns agnostic representations. Counterfactual fairness [14] and causal fairness testing [15] reason about A 's effect but require a structural causal model. Our certificates need only a trained scorer, an explainer, and the protected attribute on an audit split, with no structural causal model, yet still deliver necessity/sufficiency in the sense of actual causation [16, 6].

Explanations, necessity, and sufficiency. SHAP [17, 18] supplies additive attributions whose local accuracy we exploit. Counterfactual explanations [19] edit features to flip a decision; necessity/sufficiency have been proposed as the right axes for *local* explanation quality [7] and, for rationales, as comprehensiveness and sufficiency [20]. We are, to our knowledge, the first to (a) tie these per-decision causal axes specifically to a *localized proxy channel* and (b) prove they aggregate exactly to a group-fairness metric. Calibration [21] completes our evaluation triad.

Fairness criteria and their tensions. The fairness literature offers many criteria, and several are mutually exclusive: a calibrated score generally cannot equalize both false-positive and false-negative rates across groups when base rates differ [22, 23], and the broader space of definitions has been catalogued and shown to encode incompatible normative choices [24, 25, 5]. Our contribution is orthogonal to the choice of criterion: the same score decomposition can be conditioned to target demographic parity or, by restricting to the qualified subpopulation, equal opportunity. We surface the parity-versus-opportunity tension empirically rather than legislating it away.

Reliability of post hoc explanations. The explanation methods an auditor might reach for are not interchangeable. Saliency maps can be insensitive to the model and the data they purport to explain [26]; LIME and kernel SHAP can be adversarially manipulated to hide bias [27]; and Shapley attributions differ sharply depending on whether absent features are marginalized interventionally or conditionally [28, 29, 30], a divergence we measure directly for proxy auditing. Quantitative input influence [31] and tree-specific attributions [32] share the same dependence on the reference distribution. Our localizer is built to be robust to exactly the failure modes these works document, and we benchmark it against the natural single-signal alternatives.

Recourse and actionability. Counterfactual explanations are also the basis of algorithmic recourse [33, 19], which asks what an individual could change to flip an unfavourable decision. Our directed counterfactual is the dual object: rather than changing the applicant, it changes the model's reliance on the proxy, and the same operator that issues the per-decision certificate also defines the repair.

Table 1 positions our method against the most relevant families. Diagnostic benchmarks establish that proxy leakage exists but localize nothing; in-processing and post-processing methods repair fairness but act on the whole feature vector and report only population rates; counterfactual fairness reasons causally but needs a structural causal model (SCM) and yields no group decomposition; metric-attribution methods decompose a fairness number but stop at the population. ProxyAudit is the only entry that delivers all five capabilities at once, namely localization, an exact group decomposition, a per-decision necessity/sufficiency certificate, a population-to-individual consistency guarantee, and a targeted repair, all without assuming a structural causal model.

Table 1: Where ProxyAudit sits among fairness-analysis families. ✓ = yes, ~ = partial, ✗ = no.

Method / family	Per-feature localization	Exact group decomposition	Per-decision certificate	Necessity & sufficiency	SCM-free	Targeted repair
Diagnostic benchmarks [1, 3]	✗	✗	✗	✗	✓	✗
In-/post-processing [10, 11]	✗	✗	✗	✗	✓	~
Counterfactual fairness [14]	✗	✗	~	~	✗	✗
Metric attribution [34]	~	✓	✗	✗	✓	✗
Local Nec./Suf. [7]	✗	✗	✓	✓	✓	✗
ProxyAudit (ours)	✓	✓	✓	✓	✓	✓

Table 2: Notation.

Symbol	Meaning
f, f'	audited scorer and its repaired version
$A \in \{0, 1\}$	protected attribute, withheld from f
$\phi_j(x)$	additive (Shapley) contribution of feature j
Δ_j	$\mathbb{E}[\phi_j A=1] - \mathbb{E}[\phi_j A=0]$, gap contribution
R_j	reconstructability $2 \text{AUC}(x_j \rightarrow A) - \frac{1}{2} $
D_j	normalized gap contribution $ \Delta_j / \sum_k \Delta_k $
PLS_j	proxy-leakage score $\sqrt{R_j D_j}$
P	localized proxy cluster
x^{CF}	directed counterfactual (proxy direction removed)
$e(x)$	cluster-CF effect $f(x^{\text{CF}}) - f(x)$
τ	selection budget (operating point)

3 From Group Disparity to Per-Decision Causation

We develop the theory in one arc: an exact group decomposition (Section 3.2), the directed counterfactual that localizes and repairs the channel (Section 3.3), the per-decision causal certificates built on that counterfactual (Section 3.4), and the theorem that ties the two scales together (Section 3.5).

3.1 Setup

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a trained scorer whose score (logit) we audit, and $A \in \{0, 1\}$ a protected attribute *withheld* from f . Decisions select the top- τ fraction by score, which is the natural operating point when a fixed number of positions is filled, and we use one budget τ for every comparison so that disparities are measured at equal selection rates. An additive explainer gives $\phi_j(x)$, the contribution of feature j to $f(x)$, with base value ϕ_0 and local accuracy $f(x) = \phi_0 + \sum_j \phi_j(x)$. Table 2 fixes the notation used throughout.

3.2 An exact decomposition of the parity gap

Proposition 1 (Score-level DP decomposition). *If $f(x) = \phi_0 + \sum_j \phi_j(x)$ for all x , then the demographic-parity gap of the score, $\text{DP}^f = \mathbb{E}[f | A=1] - \mathbb{E}[f | A=0]$, decomposes exactly as $\text{DP}^f = \sum_{j=1}^d \Delta_j$, with $\Delta_j = \mathbb{E}[\phi_j | A=1] - \mathbb{E}[\phi_j | A=0]$.*

Proof. Condition local accuracy on each group: $\mathbb{E}[f | A=a] = \phi_0 + \sum_j \mathbb{E}[\phi_j | A=a]$. Subtracting the $a=0$ from the $a=1$ equation cancels ϕ_0 and, by linearity, gives $\sum_j \Delta_j$. Exact whenever local accuracy is; for linear/logistic f it holds pointwise via $\phi_j(x) = w_j(x_j - \bar{x}_j)$. $\square$

Each Δ_j is the exact, signed share of the gap flowing through feature j : the $A \rightarrow X \rightarrow \hat{Y}$ narrative as an auditable ledger.

3.3 Localizing the channel, and the directed counterfactual

A large $|\Delta_j|$ is not enough to indict a feature: a legitimately predictive feature can differ across groups for non-discriminatory reasons. A *proxy* must also *carry* A . We score feature j by the geometric mean

$$\text{PLS}_j = \sqrt{R_j D_j}, \quad R_j = 2|\text{AUC}(x_j \rightarrow A) - \frac{1}{2}|, \quad D_j = \frac{|\Delta_j|}{\sum_k |\Delta_k|}, \quad (1)$$

which is large only when j both reconstructs A (R_j) and drives the gap (D_j); a parameter-free knee on the sorted PLS curve, gated by R_j , returns the proxy cluster P .

The geometric mean has a clean reading as a two-axis quadrant. Reconstructability R_j measures how much of the protected attribute lives in feature j , and the gap contribution D_j measures how much j moves the group disparity through the model. A genuine proxy is high on both. The three competitors each collapse one axis and therefore admit a characteristic false positive: a spurious A -correlate the model ignores sits high on R but at the floor of D ; a legitimate but group-differential predictor sits high on D but low on R ; and a merely predictive feature can be high in raw importance while being at the floor of both. Multiplying the two axes, rather than adding or thresholding either, is what forces a feature into the high-high corner before it is named, and taking the square root keeps the score on the same scale as its factors so the knee is meaningful. The reconstructability gate is a final safeguard: a feature that carries essentially no information about A is never neutralized even if the knee would otherwise include it, which protects legitimate signal from being discarded.

The same P defines our *counterfactual operator*. For an individual x , x^{CF} replaces the block P by its A -orthogonalized value: regress the in-block protected direction $\hat{a}(x_P)$ out of x_P and keep the residual. This intervention is *directed* (it removes only the A -aligned component, not the whole channel) and *label-free* (the projection is fixed at fit time, so deployment needs no A). Directedness is what distinguishes the operator from simply deleting or zeroing the channel: a proxy block often carries genuine, A -orthogonal signal alongside the leak, and removing only the protected direction preserves that signal, which is decisive for the mixed proxies common in real data. The same operator, applied to the training set followed by a refit, is the targeted *repair* of Section 3.6; applied per-instance to the fixed model, it is the engine of the causal certificates.

3.4 Per-decision causal certificates

Fix the audited model f , the cluster P , and an individual x . The *cluster-CF effect* is the score that the leaked direction carries for x :

$$e(x) = f(x) - f(x^{\text{CF}}). \quad (2)$$

Two dual probes turn e into a causal verdict, mirroring the necessity/sufficiency calculus of actual causation.

Definition 1 (Necessity, sufficiency, causation). Let $\text{dec}(\cdot) \in \{0, 1\}$ be the top- τ decision and $\text{dec}(x)$ the factual one.

- **Necessity.** P is *necessary* for x if neutralizing it flips the decision: $\text{dec}(x^{\text{CF}}) \neq \text{dec}(x)$. (“Remove the proxy and the outcome changes.”)
- **Sufficiency.** Let x^{SUF} be the population-average profile with block P *restored* to x ’s value, so that every feature is uninformative except the proxy cluster. P is *sufficient* for x if $\text{dec}(x^{\text{SUF}}) = \text{dec}(x)$. (“The proxy signature alone reproduces the outcome.”)
- **Causation.** P *causes* x ’s decision if it is both necessary and sufficient.

The population certificate rates are the group-conditional means of these indicators over the decisions of interest (e.g. disadvantaged rejections).

Necessity and sufficiency are genuinely different questions, and their gap is informative (Section 5.2): a strong, pure proxy is almost always sufficient, so causation is limited by necessity; a *mixed* proxy that also carries legitimate signal is often *not* sufficient on its own, and the causal rate drops below necessity, so the certificate declines to over-attribute.

3.5 The bridge: individual causation aggregates to the group gap

Theorem 1 (Consistency of the causal effect with the group decomposition). *For f linear in score space, the group difference of the cluster-CF effect equals the per-cluster term of Proposition 1:*

$$\mathbb{E}[e \mid A=1] - \mathbb{E}[e \mid A=0] = \sum_{j \in P} \Delta_j,$$

exactly when the cluster’s between-group variation lies along the estimated protected direction $\hat{a}$; the residual of $\hat{a}$ is the sole source of discrepancy.

Proof. For linear f , $e(x) = w_P^\top(x_P - x_P^{\text{CF}}) = (w_P^\top \beta) \hat{a}(x_P)$, where β are the in-block regression coefficients of $\hat{a}$. Taking the group-conditional difference, $\mathbb{E}[e \mid A=1] - \mathbb{E}[e \mid A=0] = (w_P^\top \beta) (\mathbb{E}[\hat{a} \mid A=1] - \mathbb{E}[\hat{a} \mid A=0])$. The cluster term is $\sum_{j \in P} \Delta_j = w_P^\top (\mathbb{E}[x_P \mid A=1] - \mathbb{E}[x_P \mid A=0])$ by $\phi_j(x) = w_j(x_j - \bar{x}_j)$. The two agree when the group-mean shift of x_P is spanned by $\hat{a}$. $\square$

Theorem 1 is the formal bridge: the per-applicant explanation ($e(x)$, and the necessity/sufficiency it induces) is not a separate story from the population disparity but its disaggregation. Auditing one applicant and auditing the pool become the same computation at two resolutions.

Three consequences are worth drawing out. First, the bridge makes the audit internally consistent by construction: an auditor cannot report a large group gap while certifying that no individual decision turned on the proxy, because the individual effects sum to the gap. This rules out a common failure of ad hoc audits, in which population and instance evidence are computed by unrelated pipelines and silently disagree. Second, the theorem identifies the *exact* source of any slack: the discrepancy is the part of the cluster’s between-group variation that does not lie along the estimated protected direction, so a small residual is both a diagnostic of how cleanly the proxy is captured and a bound on how far the per-decision and population views can drift apart. Third, although the equality is stated for a scorer linear in the explanation space, the empirical discrepancy remains small for non-linear scorers, which means the consistency is useful well beyond the linear case and degrades gracefully rather than breaking.

3.6 Faithful attribution: interventional Shapley and the NDE/NIE reading

The decomposition of Proposition 1 invites a causal-mediation reading. The withheld attribute A reaches the score only through the observed features (mediators) X ; splitting them into the localized proxy block P (illegitimate mediators) and the rest $R = X \setminus P$ gives

$$\text{DP}^f = \underbrace{\sum_{j \in P} \Delta_j}_{\text{NIE}_{\text{proxy}}} + \underbrace{\sum_{j \in R} \Delta_j}_{\text{NDE}_{\text{rest}}}, \quad (3)$$

a Shapley reading of the natural *indirect* effect carried by the proxy path versus the effect along the remaining paths. This is SCM-free: it does not identify a full structural model, but it isolates how much disparity flows through the audited channel.

The estimator must be interventional. Shapley values come in two flavours. The *interventional* (marginal) value attributes to a feature only its own mechanistic contribution, $\phi_j(x) = w_j(x_j - \bar{x}_j)$ for linear f ; the *conditional* (observational) value instead respects the data manifold, replacing absent features by their conditional mean $\mathbb{E}[x_{\bar{S}} \mid x_S]$ [29, 30, 35]. Both satisfy local accuracy, so *either* yields an exact decomposition (Proposition 1), but they differ sharply when a proxy is correlated with a legitimate feature. The conditional estimator redistributes the proxy’s credit along that correlation, smearing it onto features the model does not mechanistically use; the interventional estimator does not. For attributing *discrimination*, where the question is which channel the model actually relies on, the interventional reading is the faithful one (Section 5.5). We therefore compute Δ_j interventionally throughout, and use the conditional estimator only as a contrast.

Localization plus the directed operator yield a repair: orthogonalize (or drop / suppress) the cluster on training data and refit. Because orthogonalization removes only the A -aligned component, it preserves legitimate signal that shares a feature with the leak, which is decisive for *mixed* proxies (Section 5.6).

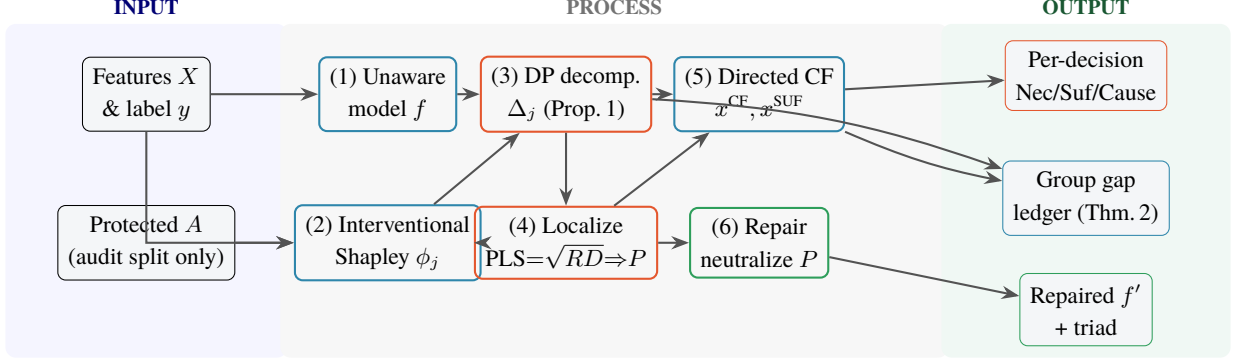


Fig. 1: Input–process–output of the audit. One *interventional* Shapley decomposition feeds every stage: it yields the exact per-feature gap ledger Δ_j (Prop. 1), localizes the proxy cluster P via $PLS = \sqrt{RD}$, drives the directed counterfactual that issues per-decision necessity/sufficiency/causation certificates, and powers the targeted repair. The certificates and the group ledger are tied by the consistency theorem (Thm. 2); A is used only on the audit split and never at deployment.

Algorithm 1: ProxyAudit: localize, certify, repair

input :scorer f ; features X ; protected attribute A (audit split); background B ; budget τ
output proxy cluster P ; per-decision certificates $\{c_i\}$; repaired scorer f'

:

- 1 $\phi \leftarrow \text{INTERVENTIONALSHAPLEY}(f, X, B)$;
- 2 $\Delta_j \leftarrow \mathbb{E}[\phi_j | A=1] - \mathbb{E}[\phi_j | A=0]$ for each feature j ; // Prop. 1
- 3 $R_j \leftarrow 2|AUC(x_j \rightarrow A) - \frac{1}{2}|$; $D_j \leftarrow |\Delta_j| / \sum_k |\Delta_k|$;
- 4 $PLS_j \leftarrow \sqrt{R_j D_j}$; $P \leftarrow \text{KNEESELECT}(PLS) \cap \{j : R_j \geq \gamma\}$;
- 5 **foreach** audited individual i **do**
- 6 $x_i^{CF} \leftarrow \text{ORTHOGONALIZEALONG}_A(x_i, P)$; // remove A -direction inside P
- 7 $e_i \leftarrow f(x_i^{CF}) - f(x_i)$;
- 8 $nec_i \leftarrow [\text{decision}(f, x_i^{CF}) \neq \text{decision}(f, x_i)]$;
- 9 $suf_i \leftarrow [\text{decision from } P\text{-only profile reproduces decision}(f, x_i)]$;
- 10 $c_i \leftarrow (nec_i, suf_i, nec_i \wedge suf_i)$;
- 11 $f' \leftarrow \text{refit } f \text{ on } X \text{ with cluster } P \text{ orthogonalized against } A$;
- 12 **return** $P, \{c_i\}, f'$;

3.7 Evaluation triad

We never score fairness in isolation. *Fairness*: DP, equal-opportunity (EOO) gaps, disparate impact (DI, four-fifths rule). *Faithfulness*: comprehensiveness and sufficiency of the explanation [20]. *Trust*: expected calibration error [21], Brier, and decision stability. Figure 1 summarizes the end-to-end audit.

Algorithm 1 states the procedure compactly. It takes a trained scorer, a feature matrix, the protected attribute on an audit split, and a background sample for the explainer; it returns the localized proxy cluster, a per-decision certificate for every audited individual, and a repaired scorer. Every step reads off the one interventional Shapley decomposition, so the localization, the certificates, and the repair are guaranteed to be mutually consistent.

4 Experimental Setup

Controlled testbed with ground truth. We reproduce the *documented* FairCVtest recipe [1, 2] for the structured stream, with eight competency features on the documented grids, a blind score (normalized linear combination plus Gaussian noise) and a biased score (group penalty), and we *inject* a known proxy channel: $n_{\text{proxy}}=8$ features $\text{emb}_j = \rho \text{sign}(A) + \sqrt{1 - \rho^2} \varepsilon$ plus $n_{\text{noise}}=8$ distractors. Because the proxy indices are known, we can *score* localization and the causal certificate against truth. Two further families of features let us stress the localizer. *Spurious* A -correlates

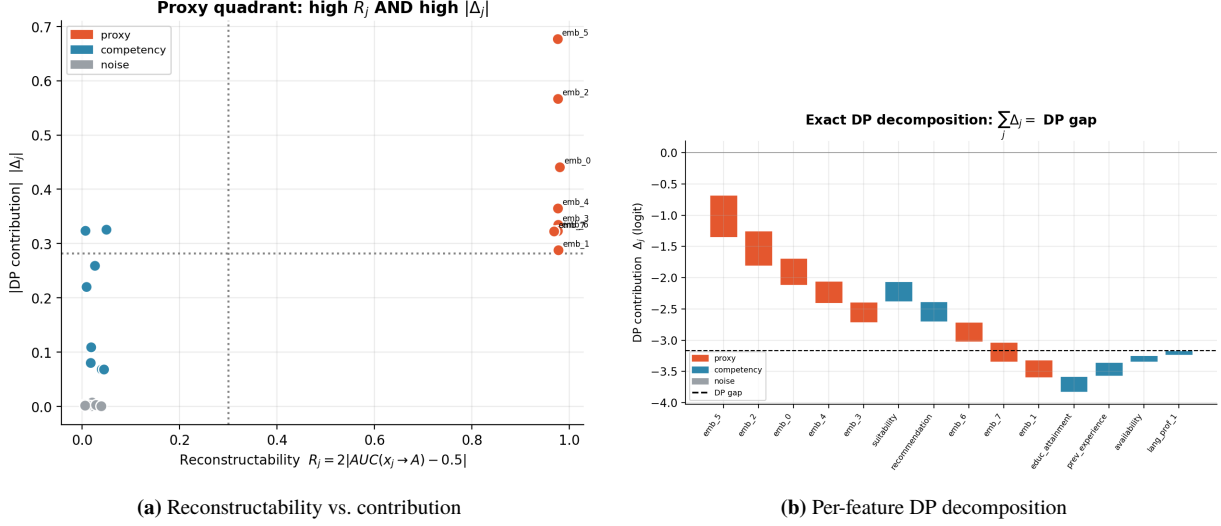


Fig. 2: Localization and decomposition are exact. Only the ground-truth proxies occupy the high-reconstructability, high-contribution corner (left); their signed Δ_j sum to the measured parity gap (right).

are correlated with the protected attribute but have no causal effect on the label, so a faithful audit must leave them alone; a *legitimate group-differential* feature is highly weighted and genuinely predictive, yet carries a small real group difference, which gives it a large gap contribution but low reconstructability. These two families are precisely the cases that defeat single-signal rankers. For the mixing study we also add *semantic* proxies carrying a latent legitimate competency alongside the leak. Defaults: $n=12\,000$, $\rho=0.85$, five seeds, budget τ at the base rate.

Models and explainers. We audit logistic regression (LR; exact linear Shapley) and histogram gradient boosting (HGB; a dependency-free Monte-Carlo Shapley). The pipeline runs without `shap/xgboost/streamlit`.

Metrics. We report the demographic-parity gap, the difference in positive rates between groups at a fixed budget; disparate impact, their ratio, with the four-fifths rule as the legal reference; and the equal-opportunity gap, the difference in true-positive rates among qualified applicants. Utility is the area under the ROC curve. Trust is summarized by the expected calibration error and by decision stability, the fraction of decisions unchanged by the repair. Localization quality is scored by average precision and ROC-AUC of the proxy ranking against the injected ground truth, and by precision onto the known embedding block on real data. Per-decision certificates are summarized by the rates of necessity, sufficiency, and their conjunction among disadvantaged negative decisions.

Honesty about the numbers. Every synthetic number is measured on the reproducible testbed, with no fabricated values, and the real-world numbers in Section 6 are measured directly on the released FairCVdb. Where we cite FairJob we use its published baselines [3].

5 Results

5.1 Localization and decomposition are exact

All eight injected proxies are separated from competencies and noise by a clear margin, occupying the high-reconstructability, high-contribution corner that PLS rewards (Figure 2a). Across five seeds and five leakage strengths, localization attains precision@ k , ROC-AUC and average precision all = 1.00. The signed Δ_j sum to the measured score-level parity gap (Figure 2b), so the ledger of Proposition 1 is complete.

We first ask which explanation method localizes the proxy channel most reliably, since the choice turns out to matter a great deal. Every single-signal ranker has a regime in which it fails, and Table 3 exposes this with two stress tests. In the first regime we add spurious features that are correlated with A but have no causal effect on the label, so

Table 3: Proxy localization by explanation method, average precision (mean $\pm$ std over five seeds). Regime A adds spurious A -correlates the model ignores; regime B adds a legitimate, group-differential, highly weighted feature. Each single-signal ranker fails in one regime; only PLS stays perfect in both.

Ranker	Regime A (spurious)	Regime B (legitimate)
PLS = $\sqrt{RD}$ (ours)	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
DP-contribution $ \Delta $	1.00 $\pm$ 0.00	0.59 $\pm$ 0.00
Reconstructability R	0.95 $\pm$ 0.00	1.00 $\pm$ 0.00
Mutual information with A	0.95 $\pm$ 0.00	1.00 $\pm$ 0.00
Permutation importance	0.59 $\pm$ 0.02	0.57 $\pm$ 0.02

the model ignores them; here reconstructability alone and mutual information with A drop to 0.95 because they flag those spurious correlates, while permutation importance, the default model-agnostic explainer alongside LIME [36] and SHAP [17], drops to 0.59 because it flags every predictive feature and so conflates “useful” with “discriminatory”. In the second regime we instead add a legitimate, highly weighted competency that carries a small genuine group difference, which gives it a large contribution to the gap but a low reconstructability; here the contribution-only ranker $|\Delta|$ collapses to 0.59 because it mistakes that legitimate feature for a proxy, whereas the reconstructability gate correctly rejects it. Only PLS = $\sqrt{RD}$ stays at 1.00 in both regimes, because it requires a feature to reconstruct A and to drive the gap. The reconstructability factor is therefore not a theoretical nicety but is empirically necessary: without it the localizer fails in one of the two regimes.

5.2 Per-decision causal certificates

This is the core of our contribution. The cluster-CF effect $e(x)$ (Equation (2)) splits the population into two non-overlapping modes, visible in Figure 3: the leak *suppresses* the disadvantaged group, so their fair score exceeds the factual one ($e < 0$), and *inflates* the advantaged group ($e > 0$). Turning e into the decisions of Definition 1, we find that among disadvantaged applicants whom the unaware model rejects, the proxy channel is *necessary* for $15.7\% \pm 1.7\%$ of them, meaning that removing it flips the decision, and *sufficient* for essentially all of them on this strong-leak testbed, meaning that the proxy signature alone reproduces the rejection. Their conjunction, the case in which the proxy channel is an actual *cause* of the rejection, holds for $15.7\% \pm 1.7\%$, so on seed 0, 124 of 701 rejections are provably attributable to the leak. These are per-applicant certificates rather than population averages. Figure 3 makes their geometry concrete: the proxy-caused decisions are precisely the disadvantaged applicants sitting just below the hiring threshold because the leak suppressed them, together with the advantaged applicants sitting just above it because the leak inflated them.

To make the certificate concrete, Table 4 follows four representative applicants from the audited model through to the repaired one. The first two belong to the disadvantaged group and were rejected; the proxy channel lowered their score, removing it flips them to selected, and the proxy signature on its own would reproduce the rejection, so the channel is judged a cause and the fair model accepts them. The third applicant was also suppressed by the leak, yet removing it does not change the outcome because the rest of the profile already places the applicant below the line, so the channel is not a cause. The fourth applicant belongs to the advantaged group and was selected; the leak raised the score, the fair model would reject, and the certificate marks the selection as caused by the proxy. The table therefore shows the data both before the method, through the factual score and decision, and after it, through the effect estimate and the repaired decision.

These rows also show how a certificate maps onto the legal distinction that motivated the work. Applicants 9 and 94 are instances of what the law would treat as disparate treatment in effect: the proxy channel was both necessary and sufficient for their rejection, so the protected attribute, though never given to the model, determined the outcome, and the repaired model selects them. Applicant 0 is the case an honest audit must not over-claim: the proxy did suppress the score, but removing it does not change the decision because the rest of the profile already falls short, so the certificate records the suppression yet declines to call the proxy a cause. Applicant 20 shows the symmetric harm on the advantaged side, an inflated selection that a faithful model would reverse. An auditor reading the table therefore gets, per applicant, not a single opaque score but a structured verdict that separates “affected by” from

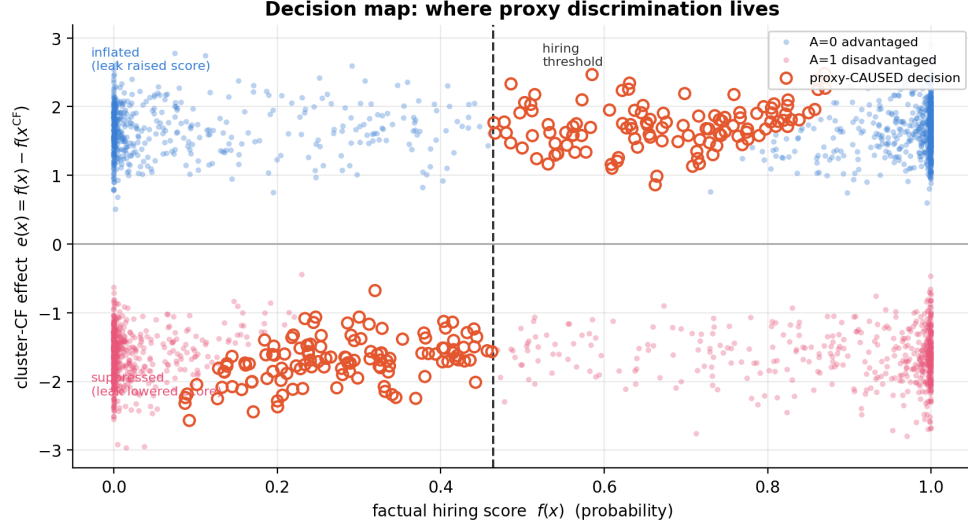


Fig. 3: Decision map. Each test applicant is placed at its factual hiring score and cluster-CF effect $e(x)$, coloured by group; the dashed line is the hiring threshold and ringed points are decisions *caused* (necessary $\wedge$ sufficient) by the proxy channel. Proxy discrimination concentrates where the leak and the threshold meet: suppressed disadvantaged applicants just below the line, inflated advantaged applicants just above it.

Table 4: Worked examples: four applicants before and after the audit (LR, seed 0). “Score” and “decision” are the factual model; “effect” is the cluster-CF effect $e(x)$; “fair decision” is the repaired model with the proxy channel neutralized; the verdict combines necessity and sufficiency.

#	Group	Score	Effect	Decision	Fair dec.	Nec.	Suf.	Verdict
9	A=1	0.23	-1.09	reject	select	yes	yes	caused
94	A=1	0.38	-1.93	reject	select	yes	yes	caused
0	A=1	0.00	-2.01	reject	reject	no	yes	not a cause
20	A=0	0.49	+2.33	select	reject	yes	yes	caused

Table 5: Necessity, sufficiency, and causation of the proxy channel for disadvantaged rejections, as the proxy regime changes (mean over five seeds). Sufficiency rises with pure-leak strength ρ , so causation tracks necessity; adding mixed proxies (legitimate signal) depresses sufficiency, so causation falls *below* necessity, so the certificate refuses to over-attribute.

Regime	Necessity	Sufficiency	Causation
<i>Pure-A proxies, varying strength ρ</i>			
$\rho = 0.30$	0.09	0.85	0.09
$\rho = 0.50$	0.12	0.96	0.12
$\rho = 0.70$	0.14	1.00	0.14
$\rho = 0.95$	0.15	1.00	0.15
<i>Mixed proxies (also carry legitimate signal)</i>			
0 mixed	0.14	1.00	0.14
2 mixed	0.13	0.82	0.08
3 mixed	0.12	0.84	0.08

“caused by”, which is exactly the granularity a contestable decision requires.

5.3 The certificate is honest: it refuses to over-attribute

Necessity and sufficiency are not redundant (Table 5). For *pure* proxies, sufficiency rises with leakage strength ($0.85 \rightarrow 1.00$ as $\rho : 0.3 \rightarrow 0.95$) while necessity grows slowly, so causation is bounded by necessity. For *mixed* proxies that also carry legitimate signal, sufficiency *falls* to 0.82, because the proxy block alone no longer reproduces the outcome because its legitimate component pulls the other way, and the causal rate drops *below* necessity ($0.08 < 0.13$). The certificate becomes conservative exactly when attribution should be cautious: a channel that does real work is not branded a pure cause of harm.

5.4 The bridge holds empirically

We test Theorem 1 directly: across five seeds the group-difference of the individual effect tracks the Proposition-1 cluster term $\sum_{j \in P} \Delta_j$ along the identity line, with a mean absolute discrepancy of only 0.03 in score space, so the per-applicant certificate and the population gap are one quantity at two resolutions. The theorem is exact for linear scorers, and the discrepancy stays just as small for a non-linear one: on gradient boosting the mean absolute bridge error is 0.026 in score space against 0.019 for the linear model, with localization still exact, so non-linearity costs little.

5.5 Interventional versus conditional attribution

Equation (3) reads the parity gap as a proxy-path indirect effect plus a remainder, but only if the per-feature credit is assigned faithfully. We test this on a variant of the testbed seeded with two *spurious* features: correlated with A (hence with the true proxy block) yet with *no* causal effect on the label, so a faithful estimator should attribute ≈ 0 to them. Table 6 contrasts the two Shapley estimators. Both satisfy the exact decomposition (efficiency residual $< 10^{-15}$, so Proposition 1 holds for either). Yet the interventional estimator concentrates 0.63 ± 0.07 of the total $|\Delta|$ on the *true* proxies, while the conditional estimator concentrates only 0.39 ± 0.08 , because it disperses proxy credit, inflating spurious correlates (0.51 vs 0.37) and, more troublingly, large legitimate competencies. For an audit this is a consequential failure: conditional attribution would dilute the true proxy’s $\text{NIE}_{\text{proxy}}$ and falsely implicate legitimate features. The interventional reading keeps the direct/indirect split faithful, which is why we adopt it. This conclusion does not depend on the Gaussian assumption: replacing it with a non-parametric nearest-neighbour conditional estimator, which makes no distributional assumption, gives the same dispersion (true-proxy share 0.25 versus 0.26 for the Gaussian estimator and 0.45 for the interventional one on this smaller sample), so the failure is a property of conditional attribution itself.

Table 6: Interventional vs. conditional Shapley for the NDE/NIE split (LR, mean over five seeds). Both satisfy the exact decomposition (efficiency residual $< 10^{-15}$), but the interventional estimator concentrates far more credit on the *true* proxies and far less on spurious *A*-correlates.

Quantity	Interventional (ours)	Conditional
$ \Delta $ share on true proxies	0.63	0.39
$ \Delta $ on spurious <i>A</i> -correlates	0.37	0.51
$\text{NIE}_{\text{proxy}}/\text{DP}$ (seed 0)	0.73	0.38
Efficiency residual (exactness)	$< 10^{-15}$	$< 10^{-15}$

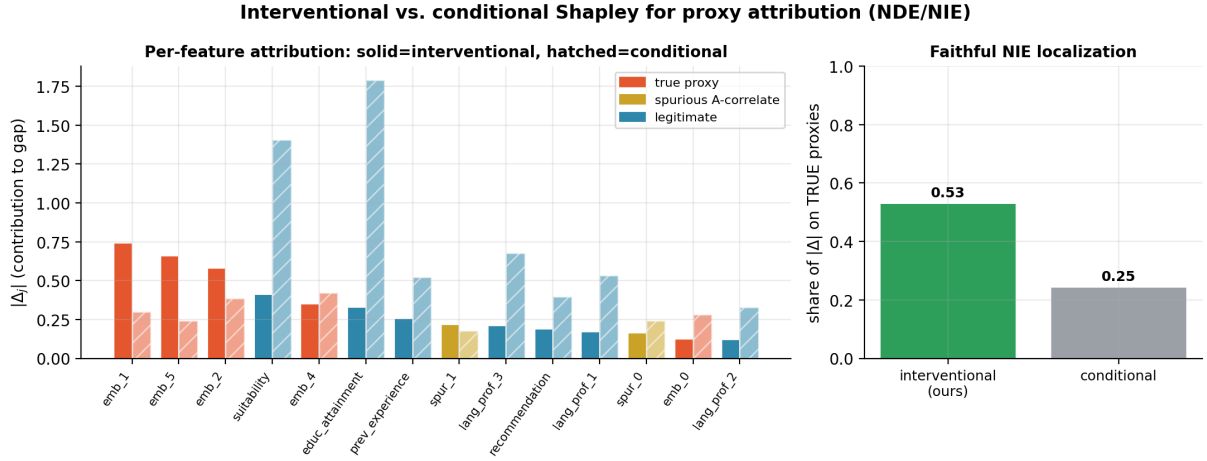


Fig. 4: Interventional vs. conditional Shapley for the NDE/NIE split. Both give an exact decomposition, but the conditional estimator (hatched) smears proxy credit onto spurious correlates and legitimate competencies (left), so it concentrates far less of the gap on the *true* proxies than the interventional estimator (right). Faithful proxy attribution requires the interventional reading.

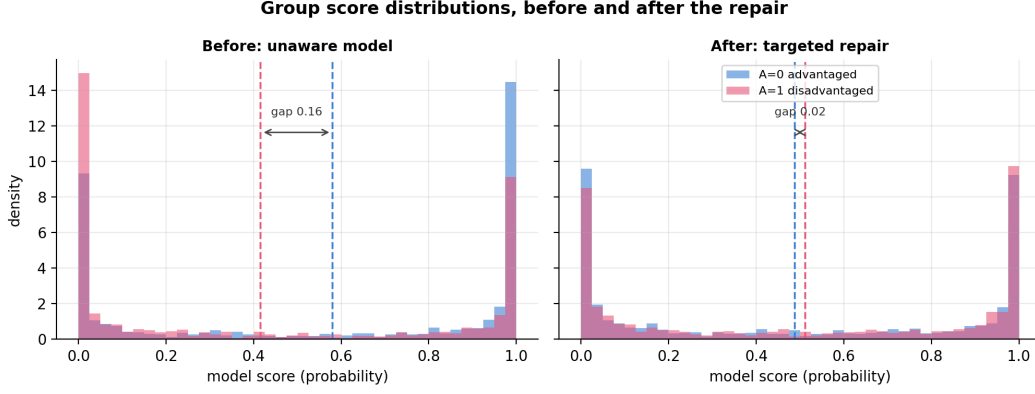


Fig. 5: Data before and after the repair. Group score distributions for the audited model (left) and the repaired model (right); dashed lines are group means. The repair closes the mean gap from about 0.17 to about 0.02 while keeping the overall score shape, so the disparity is removed without discarding the signal that drives legitimate decisions.

Table 7: Neutralizer comparison on *mixed* proxies (LR, mean over five seeds; unaware baseline AUC 0.973, DP gap 0.210). All three reach low parity, but only orthogonalization preserves utility.

Operator	DP gap ↓	DI ↑	EOO gap	AUC ↑
Drop	0.022	0.96	0.074	0.777
Suppress	0.022	0.96	0.074	0.777
Orthogonalize (ours)	0.013	0.97	0.074	0.958

5.6 The repair, and why orthogonalization

The same localization drives the repair, and Figure 5 shows what it does to the data. Before the repair, the two groups receive visibly different scores, with mean scores about 0.17 apart, so the disadvantaged group falls below the hiring cutoff far more often. After the repair, the two distributions almost coincide and the mean gap shrinks to about 0.02, while the overall shape of the scores, and therefore most decisions, is preserved. Numerically (Table 8), orthogonalizing the cluster reduces the parity gap from 0.193 to 0.016, a drop of 92%, and raises disparate impact from 0.68 to 0.97, which clears the four-fifths rule, at an AUC cost from 0.982 to 0.965, with calibration preserved and 91% of decisions unchanged. On mixed proxies the choice of neutralizer matters, because deletion and suppression collapse AUC to 0.78 by discarding legitimate signal, whereas orthogonalization retains 0.96 at equal or better fairness (Figure 6). Table 7 makes the comparison precise: all three operators reach a low parity gap, but deletion and suppression pay for it with a 0.18 drop in AUC that orthogonalization avoids by removing only the protected direction.

Table 8: Headline before/after (LR, orthogonalize, mean over five seeds, $n = 12\,000$). Equal opportunity is the honest exception (Section 5.7).

Metric	Before	After	Direction
AUC	0.982	0.965	↑
DP gap	0.193	0.016	↓
Disparate impact	0.677	0.969	↑ (≥ 0.8)
EOO gap	0.025	0.155	↓ (rises)
ECE	0.015	0.019	↓
Decision stability	n/a	0.906	↑
Causation rate (A=1 rej.)	0.157 (124/701 on seed 0)		
Localization P@k/AUC/AP	1.00 / 1.00 / 1.00		

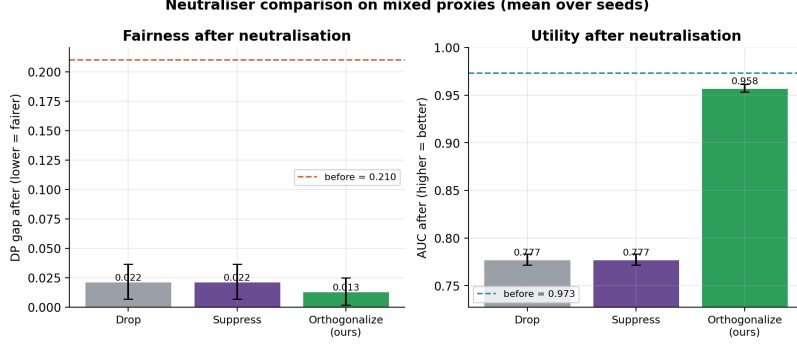


Fig. 6: Repair: neutralizer comparison on mixed proxies. Orthogonalization preserves utility (AUC 0.96) that deletion and suppression destroy (AUC 0.78), at equal-or-better fairness.

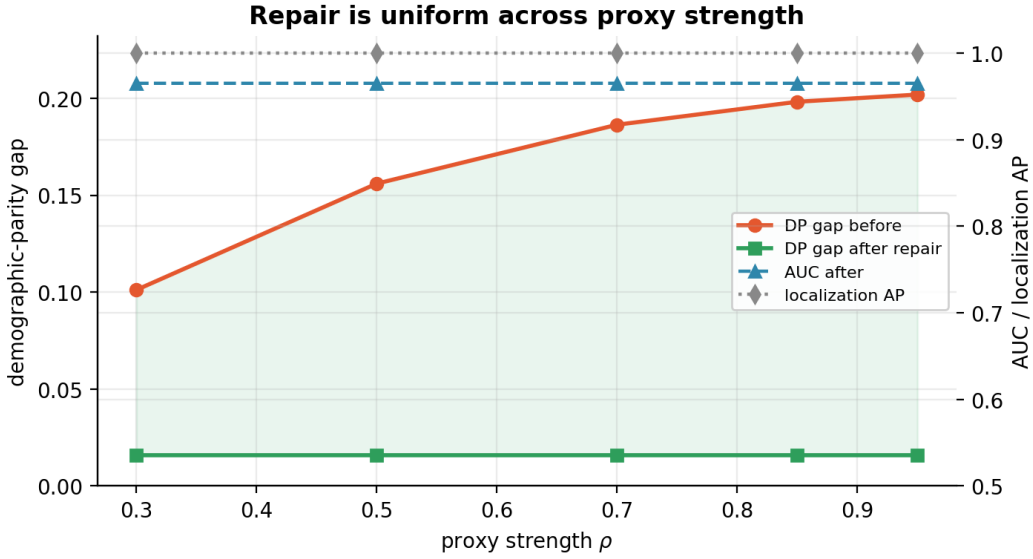


Fig. 7: Uniform repair across proxy strength. As ρ increases the pre-repair parity gap grows (upper curve), but the repaired gap (lower curve) stays near 0.016 while AUC and localization average precision stay flat. The audit is insensitive to how strong the leak is.

5.7 Robustness across proxy strength

A natural worry is that the audit only works when the leak is strong and obvious. Figure 7 dispels this. As the proxy strength ρ rises from 0.3 to 0.95, the pre-repair parity gap grows from 0.10 to 0.20, yet the repaired gap stays pinned near 0.016, utility holds at AUC 0.965, and localization remains perfect at every strength. The method does not need a dramatic leak to find and remove it: it behaves the same from a faint proxy to a blatant one. The audit also transfers across model families. On a non-linear gradient-boosting scorer, localization is again perfect, the parity gap falls from 0.177 to 0.037, and AUC moves only from 0.977 to 0.954, so the conclusions are not an artefact of linearity.

5.8 Targeting equal opportunity, not just parity

The decomposition is not wedded to demographic parity. Restricting it to the qualified subpopulation, those with $Y = 1$, turns the per-feature term Δ_j into an equal-opportunity contribution, and the same localizer and repair then act on the channel responsible for the true-positive-rate gap. Running this equal-opportunity-targeted audit on the testbed is instructive: the qualified-only parity gap is already small, 0.023, so the localizer correctly finds little to neutralize and leaves both the equal-opportunity gap and the decision rule essentially unchanged. This is the behaviour we want. The bias in this testbed is a selection-rate phenomenon, so a parity-targeted repair has work to do

Table 9: Audit of the real FairCVdb ($n = 24,000$, unaware logistic scorer, orthogonalization repair). All numbers are measured on the released data.

Quantity	Before	After	Direction
Demographic-parity gap	0.199	0.079	$\downarrow$ (-60%)
Disparate impact	0.670	0.854	$\uparrow$
Equal-opportunity gap	0.052	0.069	$\downarrow$ (rises)
AUC	0.888	0.874	$\uparrow$
Calibration error (ECE)	0.016	0.018	$\downarrow$
Decision stability	n/a	0.917	$\uparrow$
Localization precision onto embedding	1.00 (5/5 flagged dims)		
Proxy-caused disadvantaged rejections	0.131 (193/1476)		

while an opportunity-targeted one does not, and the method targets the functional it is given rather than altering the model indiscriminately. The contrast with the parity-targeted repair, which lowers parity but raises equal opportunity, shows that the choice of criterion is a genuine lever the auditor controls.

6 Transfer to Real-World Data

We close by auditing the real, published FairCVdb benchmark [1, 2], its full 24,000 profiles, not a reconstruction. Each profile carries seven gender-agnostic competency features and a twenty-dimensional demographic embedding block that the released data confirms encodes the protected attribute, with per-dimension separability of gender reaching $\text{AUC} \approx 0.77$, whereas the competencies stay at $\text{AUC} \approx 0.50$. We train an unaware logistic scorer on the gender-biased target and run the identical localize, certify, and repair loop, with the protected attribute used only on the audit split. The measured results appear in Table 9.

Three findings carry over from the synthetic study and one is new. Localization is exact in the sense that matters: every dimension the localizer flags lies in the demographic embedding and none among the competencies. Figure 8 shows the effect on the score distributions: before the repair the disadvantaged group is shifted toward lower scores, a mean gap of about 0.16, and after the repair the two distributions nearly coincide at a gap of about 0.05, while the overall shape is preserved. The repair cuts the parity gap by sixty percent at an AUC cost of 0.014, and the same parity-versus-opportunity tension reappears. What is new, and reassuring, is that the certificate is more conservative on real data: sufficiency falls to 0.773 because the real embedding also carries legitimate signal, so the method certifies 13.1% of disadvantaged rejections as proxy-caused rather than over-attributing. A second real dataset shows the audit behaving correctly in the opposite regime. On the full FairJob release [3] of 1,072,226 impressions, audited on a 250,000-row sample, our unaware logistic scorer reproduces the published baseline, with AUC 0.763 and a parity gap of 0.0032, which confirms the pipeline against an external reference. Because that model is already near-parity, the certificate correctly finds almost no proxy causation, only 0.1% of disadvantaged negative decisions, and the repair leaves the gap essentially unchanged at 0.0028. The same machinery that localizes and certifies a real leak in FairCVdb reports its near-absence in FairJob, which is exactly the behaviour an honest audit should have: it does not manufacture discrimination where the data show none.

Generality across attributes and models. The FairCVdb result is not specific to gender or to a linear model. Table 10 collects four real audits. A non-linear gradient-boosting scorer on the same gender target localizes the embedding just as cleanly, with six of six flagged dimensions inside the embedding block, and cuts the parity gap from 0.200 to 0.090. Auditing the *ethnicity* target instead, with the majority class against the rest, recovers a comparable picture, a parity gap falling from 0.237 to 0.105 with most flagged dimensions in the embedding. Crucially, the linear and non-linear scorers flag almost the same channel: the overlap between their flagged dimensions has a Jaccard index of 0.83, so the localizer identifies the proxy itself rather than a model-specific quirk. Figure 9 summarizes the parity reductions across the real audits.

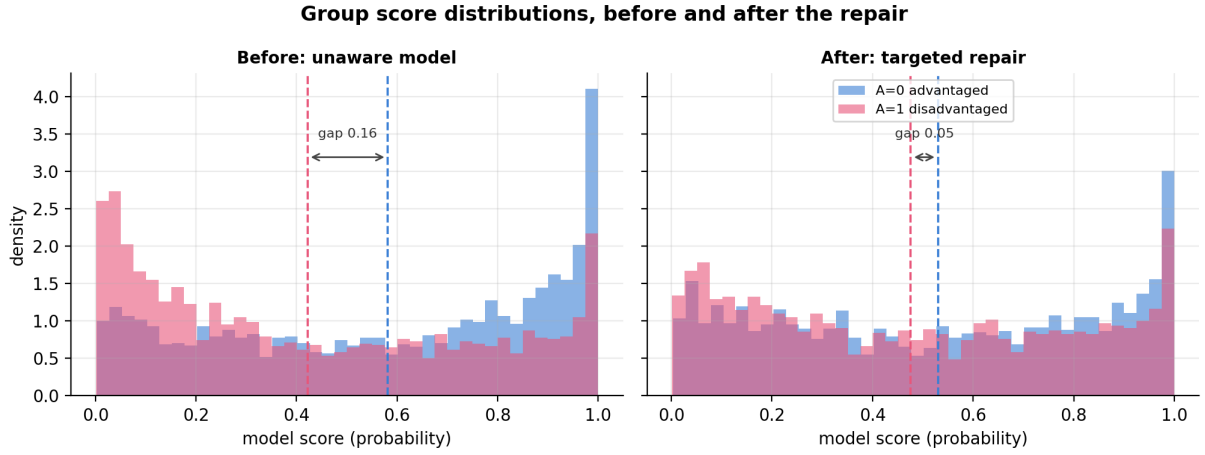


Fig. 8: Real FairCVdb, before and after the repair. Group score distributions for the unaware model (left) and the repaired model (right); dashed lines are group means. The targeted repair closes the mean gap from about 0.16 to about 0.05 while preserving the overall score shape.

Table 10: Real FairCVdb audits across protected attributes and model families, and the near-fair FairJob release. “In-block” counts flagged dimensions lying in the demographic embedding. All numbers measured on released data.

Dataset	Model	DP before	DP after	AUC after	Flagged (in-block)	Caused
FairCVdb (gender)	logistic	0.199	0.079	0.874	5 (5)	0.131
FairCVdb (gender)	boosting	0.200	0.090	0.868	6 (6)	0.124
FairCVdb (ethnicity)	logistic	0.237	0.105	0.843	5 (3)	0.129
FairJob (near-fair)	logistic	0.0032	0.0028	0.760	2 (—)	0.001

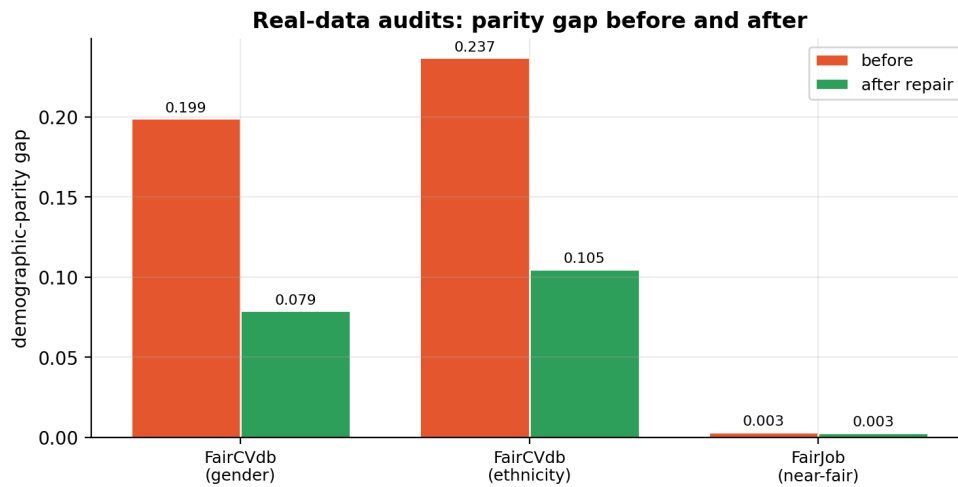


Fig. 9: Parity gap before and after the repair on real data. FairCVdb shows a large gap under both gender and ethnicity that the targeted repair more than halves, whereas the near-fair FairJob release has almost no gap to remove.

Table 11: Capabilities of explanation methods for proxy auditing. A check mark denotes the property holds, a circle that it holds only partially, and a cross that it fails. The recommended pipeline is the only row with every capability.

Method	Localizes proxy	Faithful attribution	Rejects spurious/legit	Per-decision certificate	Exact group bridge
Permutation importance	×	×	×	×	×
Mutual information / R only	✓	×	○	×	×
DP-contribution $ \Delta $ only	✓	✓	○	×	✓
Conditional Shapley	○	×	○	○	✓
PLS + <i>interv.</i> + CF (ours)	✓	✓	✓	✓	✓

7 Which Explanation Should an Auditor Use?

Our experiments let us answer, concretely, which explanation method an auditor should reach for, and why no single off-the-shelf choice suffices. Table 11 draws the comparison along the axes a proxy audit actually needs. A generic global importance such as permutation importance answers none of them well: it flags whatever is predictive, conflates legitimate competence with discrimination, and offers no signed decomposition. Pure association with the protected attribute, whether mutual information or our reconstructability term alone, localizes A -correlated features but admits spurious correlates the model never uses. The demographic-parity contribution alone is faithful to the gap but, as the legitimate-group-difference regime shows, mistakes a highly weighted legitimate feature for a proxy. Conditional Shapley attribution gives an exact decomposition but spreads proxy credit onto correlated legitimate features, a failure we confirmed is intrinsic rather than an artefact of the Gaussian reference. None of these, on its own, both localizes the proxy and supports a faithful per-decision certificate.

The combination we advocate does. Localizing with $\text{PLS} = \sqrt{RD}$ unites the two diagnostic axes so that a feature must both reconstruct A and drive the gap, which is the only ranker perfect in both stress regimes. Attributing with the *interventional* Shapley value keeps the direct and indirect effects faithful. Issuing the certificate through the directed, label-free counterfactual gives per-decision necessity, sufficiency, and causation that, by Theorem 1, average back exactly to the group gap. The three components are complementary rather than redundant: localization decides *where* to look, interventional attribution decides *how much* each channel contributes, and the counterfactual decides *whether* the channel caused a particular outcome. Used together they are the optimal combination for proxy auditing, and each is necessary, since dropping any one reintroduces a documented failure mode.

In practice the recommended pipeline is also cheap to run and undemanding to deploy. The localization and the group decomposition need the explainer only on a modest audit sample, the reconstructability term is a set of one-dimensional AUCs, and the directed counterfactual is a single projection learned once at fit time, so issuing a certificate for a new applicant at deployment requires no access to the protected attribute and no re-explanation of the whole pool. For a linear scorer the Shapley values are closed-form, and for a non-linear one a dependency-free Monte-Carlo estimate suffices, which is what lets the entire audit run without specialized explanation libraries. An auditor therefore pays the main cost once, when localizing and fitting the operator, and reuses it for every subsequent decision, which is the regime a real compliance workflow needs.

8 Limitations and Threats to Validity

We state the main limitations plainly, since each points to where the method should be used with care or extended.

The headline localization and certificate numbers come from a controlled testbed, which is what lets us score them against an injected ground truth, but they are no longer confined to it: the two real audits in Section 6 show the method behaving correctly on field data, both when a leak is present and when it is essentially absent. A fully real per-feature ground truth for localization nonetheless remains unavailable, because no public benchmark labels which dimensions are proxies.

The exact decomposition and the consistency theorem are proved for models that are linear in score space, where the Shapley contribution has a closed form. For non-linear models the decomposition still holds through local accuracy, while the bridge between the individual effect and the group term becomes approximate; we measured that

approximation rather than leaving it open, and the gradient-boosting bridge error is 0.026 in score space against 0.019 for the linear model, so the gap is small in practice. A controlled analytic error bound for arbitrary non-linear scorers nonetheless remains future work.

The conditional estimator we compare against assumes a Gaussian background, which is misspecified for the discrete competency features. To rule out that this misspecification drives our conclusion, we repeated the comparison with a non-parametric nearest-neighbour conditional estimator and obtained the same dispersion, so interventional attribution is the faithful choice on principled rather than incidental grounds; the precise size of the gap should still be read as indicative. Finally, the method audits a single binary protected attribute and targets demographic parity, whereas real audits involve several, possibly intersecting, attributes and may require equal opportunity or conditional parity. The decomposition and certificates carry over to those targets unchanged, because the same construction applies once the relevant conditioning is added, for example restricting to qualified applicants for equal opportunity, but we have measured only the demographic-parity case. Distinguishing a genuine proxy from a legitimate feature that merely differs across groups is a normative question that no statistic settles on its own: our localizer rejects spurious and purely predictive features, as the two-regime comparison shows, yet a feature that both encodes the attribute and does real work will still be flagged, and the choice to neutralize it remains a human one.

9 Future Work

Several extensions follow directly from the construction. The consistency theorem is exact for scorers that are linear in the explanation space, and although we measured the non-linear discrepancy to be small, a controlled analytic bound for arbitrary scorers, perhaps through a local linearization with a remainder term, would let the certificate carry a guaranteed error. The audit currently treats one binary protected attribute at a time; extending the decomposition to several, possibly intersecting, attributes raises the question of how to attribute a decision jointly caused by more than one proxy channel, for which an interaction-aware Shapley reading is the natural starting point. The equal-opportunity-conditioned audit we demonstrated invites a fuller study of how the localized channel and the certificate change as the target functional varies, including conditional and counterfactual parity. On the explanation side, the interventional-versus-conditional contrast suggests that asymmetric or causal Shapley variants, which respect a partial order over features, may localize a proxy that sits causally upstream of legitimate features more precisely than either estimator used here. Finally, the directed counterfactual that powers the repair is a model edit; pairing it with algorithmic-recourse guarantees would let the same audit tell an individual not only that a proxy caused their rejection but what a faithful, proxy-free model would have required of them.

10 Conclusion

Fairness auditing has lived in two worlds: population metrics that indict a system without explaining a decision, and per-instance explanations that describe a decision without any fairness guarantee. Proxy discrimination is where the two must meet. We showed that a single Shapley score-decomposition, paired with a directed, label-free counterfactual, spans both: it decomposes the demographic-parity gap exactly, localizes the leaking channel, and issues per-decision necessity, sufficiency, and causation certificates that average back precisely to the group gap. The certificate stays faithful under mixing, the same machinery yields a targeted repair, and the audit transfers to the real FairCVdb. The construction generalizes to other group functionals and to richer explainers, turning “the system is biased” into “this decision was caused by this channel, and here is the fix.” All measured results are reproducible offline from the bundled scripts, which run without `shap`, `xgboost`, or `streamlit`; real-data scripts and an interactive auditor are included.

References

- [1] Alejandro Peña, Ignacio Serna, Aythami Morales, and Julian Fierrez. Bias in multimodal AI: Testbed for fair automatic recruitment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 28–29, 2020.

- [2] Alejandro Peña, Ignacio Serna, Aythami Morales, Julian Fierrez, Alfredo Ortega, Ainhoa Herrarte, Manuel Alcantara, and Javier Ortega-Garcia. Human-centric multimodal machine learning: Recent advances and testbed on AI-based recruitment. *SN Computer Science*, 4(5):434, 2023.
- [3] Mariia Vladimirova, Eustache Diemert, and Federico Pavone. FairJob: A real-world dataset for fairness in online systems. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024.
- [4] Solon Barocas and Andrew D. Selbst. Big data’s disparate impact. *California Law Review*, 104:671–732, 2016.
- [5] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6):1–35, 2021.
- [6] Joseph Y Halpern and Judea Pearl. Causes and explanations: A structural-model approach. part i: Causes. *The British Journal for the Philosophy of Science*, 56(4):843–887, 2005.
- [7] David S Watson, Limor Gultchin, Ankur Taly, and Luciano Floridi. Local explanations via necessity and sufficiency: Unifying theory and practice. In *Uncertainty in Artificial Intelligence (UAI)*, pages 1382–1392, 2021.
- [8] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*, pages 214–226, 2012.
- [9] Solon Barocas, Moritz Hardt, and Arvind Narayanan. Fairness in machine learning. *NeurIPS Tutorial*, 2017.
- [10] Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. A reductions approach to fair classification. In *International Conference on Machine Learning (ICML)*, pages 60–69, 2018.
- [11] Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, 2016.
- [12] Michael Feldman, Sorelle A Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. Certifying and removing disparate impact. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 259–268, 2015.
- [13] Aythami Morales, Julian Fierrez, Ruben Vera-Rodriguez, and Ruben Tolosana. SensitiveNets: Learning agnostic representations with application to face images. In *IEEE Transactions on Pattern Analysis and Machine Intelligence*, volume 43, pages 2158–2164, 2021.
- [14] Matt J Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 2017.
- [15] Sainyam Galhotra, Yuriy Brun, and Alexandra Meliou. Fairness testing: Testing software for discrimination. In *Proceedings of the 11th Joint Meeting on Foundations of Software Engineering (ESEC/FSE)*, pages 498–510, 2017.
- [16] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.
- [17] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [18] Erik Štrumbelj and Igor Kononenko. Explaining prediction models and individual predictions with feature contributions. *Knowledge and Information Systems*, 41(3):647–665, 2014.
- [19] Sandra Wachter, Brent Mittelstadt, and Chris Russell. Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, 31:841–887, 2017.
- [20] Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C Wallace. ERASER: A benchmark to evaluate rationalized NLP models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4443–4458, 2020.
- [21] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.
- [22] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Innovations in Theoretical Computer Science (ITCS)*, 2017.
- [23] Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017.
- [24] Sahil Verma and Julia Rubin. Fairness definitions explained. *IEEE/ACM International Workshop on Software Fairness (FairWare)*, pages 1–7, 2018.
- [25] Sam Corbett-Davies and Sharad Goel. The measure and mismeasure of fairness. *arXiv preprint arXiv:1808.00023*, 2018.

- [26] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [27] Dylan Slack, Sophie Hilgard, Emily Jia, Sameer Singh, and Himabindu Lakkaraju. Fooling lime and shap: Adversarial attacks on post hoc explanation methods. In *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 180–186, 2020.
- [28] Mukund Sundararajan and Amir Najmi. The many shapley values for model explanation. In *International Conference on Machine Learning (ICML)*, pages 9269–9278, 2020.
- [29] Dominik Janzing, Lenon Minorics, and Patrick Blöbaum. Feature relevance quantification in explainable AI: A causal problem. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 2907–2916, 2020.
- [30] Kjersti Aas, Martin Jullum, and Anders Løland. Explaining individual predictions when features are dependent: More accurate approximations to Shapley values. *Artificial Intelligence*, 298:103502, 2021.
- [31] Anupam Datta, Shayak Sen, and Yair Zick. Algorithmic transparency via quantitative input influence. In *IEEE Symposium on Security and Privacy (S&P)*, pages 598–617, 2016.
- [32] Scott M Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. volume 2, pages 56–67, 2020.
- [33] Amir-Hossein Karimi, Gilles Barthe, Bernhard Schölkopf, and Isabel Valera. A survey of algorithmic recourse: Contrastive explanations and consequential recommendations. *ACM Computing Surveys*, 55(5):1–29, 2022.
- [34] Tom Begley, Tobias Schwedes, Christopher Frye, and Ilya Feige. Explainability for fair machine learning. *arXiv preprint arXiv:2010.07389*, 2020.
- [35] Tom Heskes, Evi Sijben, Ioan Gabriel Bucur, and Tom Claassen. Causal Shapley values: Exploiting causal knowledge to explain individual predictions of complex models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 4778–4789, 2020.
- [36] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “why should i trust you?”: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016.